

Class 9: Candy Mini-Project

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Background and Learning Objectives

In this mini-project, you will explore FiveThirtyEight’s Halloween Candy dataset. FiveThirtyEight, sometimes rendered as just 538, is an American website that focuses mostly on opinion poll analysis, politics, economics, and sports blogging. Your task is to explore their candy dataset to find out answers to these types of questions.

- Import and explore an unfamiliar dataset to quickly characterize its structure and identify variables requiring special handling;
- Create and iteratively refine effective visualizations including ranked bar plots and scatter plots with `ggrepel()` labels, and interactive `plotly()` graphics to explore and communicate patterns in data;
- Generate and interpret correlation matrices to identify relationships between variables;

- Conduct principal component analysis (PCA) with appropriate scaling, interpret variance explained, and create score plots (PC1 vs PC2) to reveal clustering structure in high-dimensional data;
- Interpret PCA loadings to understand which original variables drive separation along principal components, connecting these back to the correlation structure;
- Synthesize multiple analyses (exploration, correlation, PCA) to draw substantive, data-driven conclusions.

Data Import

```
# Import csv file with candy dataset
candy_file <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-rankings.csv"

# Read the csv file into the dataframe and use the candy names as row names
candy <- read.csv(candy_file, row.names=1)

# Show first 6 rows of dataset using `head()`
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1		0	0			1
3 Musketeers	1	0	0		0	1			0
One dime	0	0	0		0	0			0
One quarter	0	0	0		0	0			0
Air Heads	0	1	0		0	0			0
Almond Joy	1	0	0		1	0			0

	hard	bar	pluribus	sugar	percent	price	percent	win	percent
100 Grand	0	1	0	0.732	0.860	66.97173			
3 Musketeers	0	1	0	0.604	0.511	67.60294			
One dime	0	0	0	0.011	0.116	32.26109			
One quarter	0	0	0	0.011	0.511	46.11650			
Air Heads	0	0	0	0.906	0.511	52.34146			
Almond Joy	0	1	0	0.465	0.767	50.34755			

What is in the dataset?

The dataset includes all sorts of information about different kinds of candy. For example, is a candy chocolate? Does it have nougat? How does its cost compare to other candies? How many people prefer one candy over another?

Q1. How many different candy types are in this dataset?

```
# Use nrow() to count number of rows to see how many different candies there are  
nrow(candy)
```

```
[1] 85
```

There are 85 different candy types in this dataset.

Q2. How many fruity candy types are in the dataset?

```
# Obtain sum of the "fruity" column to find out how many fruity candy types are in the dataset  
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in the dataset.

What is your favorite candy?

Winpercent for a given candy is the percentage of people who prefer this candy over another randomly chosen candy from the dataset. Higher values indicate a more popular candy.

We can find the winpercent value for Almond Joy by using its name to access the corresponding row of the dataset. This is because the dataset has each candy name as rownames

```
# Find the winpercent value for Almond Joy by using its name to access the corresponding row  
candy["Almond Joy", ]$winpercent
```

```
[1] 50.34755
```

Alternate way to find winpercent using dplyr package

```
# Load dplyr package  
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
# Use dplyr functions `filter()` and `select()` to find Almond Joy winpercent by choosing the candy |>
  filter(row.names(candy)=="Almond Joy") |>
  select(winpercent)
```

```
      winpercent
Almond Joy  50.34755
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

My favorite candy is Almond joy. Its winpercent value is 50.34755.

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

The winpercent value for “Kit Kat” is 76.7686.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

The winpercent value for “Tootsie Roll Snack Bars” is 46.6535.

Side-note: the `skimr::skim()` function

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset.

```
# Install skimr packages
#install.packages("skimr")
```

```
# Instead of loading the entire library, extract one function, `skim()`, from the skimr package
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Winpercent seems to be on a different scale than the majority of the other columns in the dataset. The values seen in the winpercent row are the only values not between 1 and 0. They are on a scale of 0-100.

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

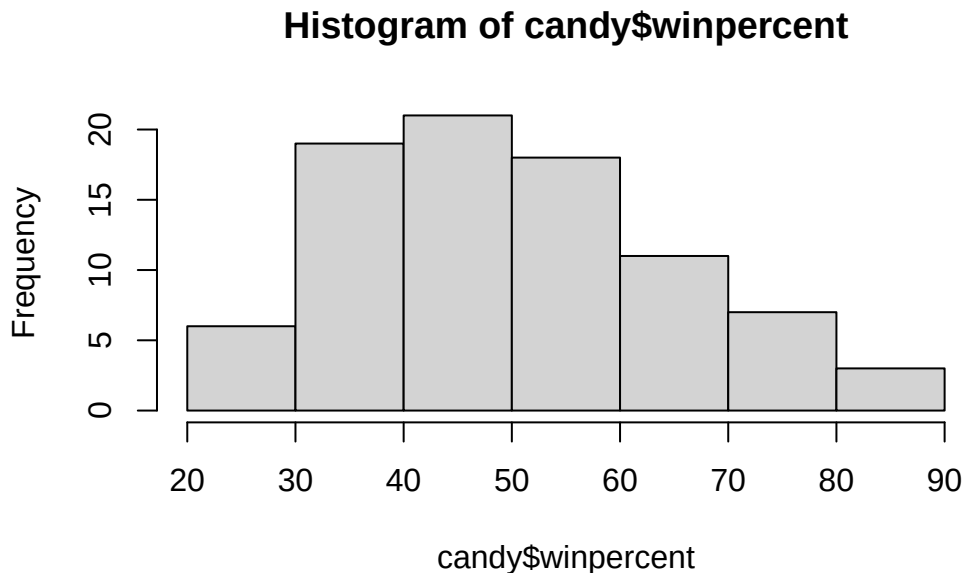
A zero means the candy does not contain chocolate and a 1 means the candy does contain chocolate.

Exploratory analysis

A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`.

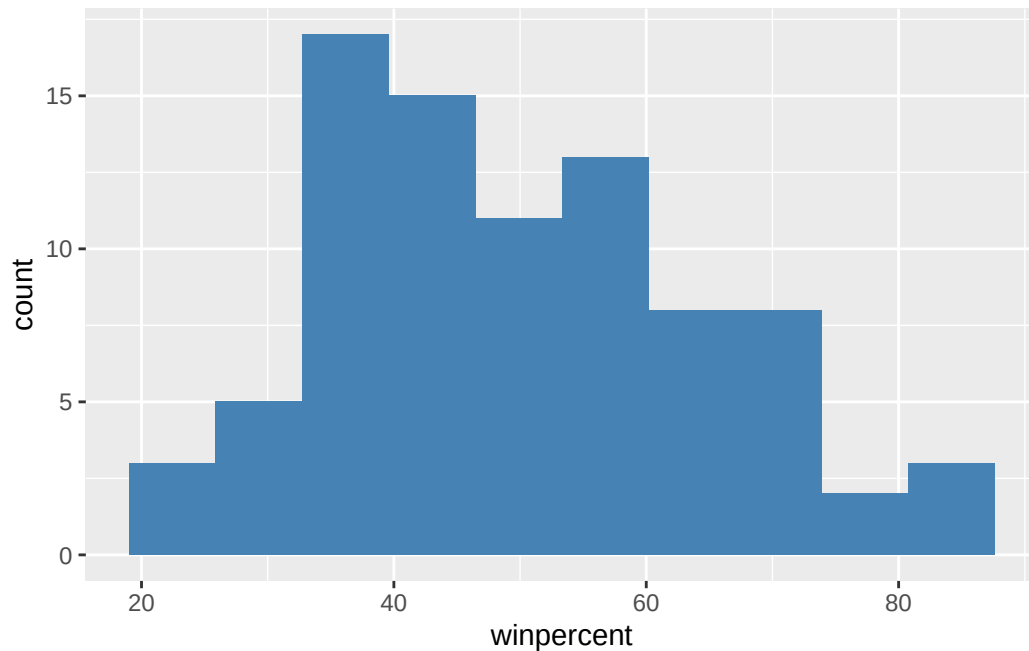
Q8 Plot a histogram of `winpercent` values using both base R and `ggplot2`

```
# Plot histogram of winpercent values using base R
hist(candy$winpercent)
```



```
# Import ggplot2 library
library(ggplot2)

# Use ggplot to create histogram of winpercent values
ggplot(candy)+
  aes(winpercent)+
  geom_histogram(bins=10, fill="steelblue")
```



```
# Use `mean()` function to find mean of winpercent values  
mean(candy$winpercent)
```

```
[1] 50.31676
```

Median is better measure of centrality

```
# Use `median()` function to find mean of winpercent values  
median(candy$winpercent)
```

```
[1] 47.82975
```

Q9 Is the distribution of winpercent values symmetrical?

No, the distribution of winpercent values is not symmetrical.

Q10 Is the center of the distribution above or below 50%?

```
# Use `summary()` function to obtain values such as median to determine centrality  
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The center of distribution is below 50%, as seen in the median which is 47.83.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# Turn "chocolate" feature into logical (a.k.a. TRUE/FALSE) values with the as.logical() function
choc.ind  <- as.logical(candy$chocolate)

# Use logical vector to access chocolate candy rows
choc.candy <- candy[choc.ind, ]

# Extract winpercent values for chocolate candy
choc.win   <- choc.candy$winpercent

# Find mean of winpercent values for chocolate candy
mean(choc.win)
```

```
[1] 60.92153
```

```
# Use same method to find mean of winpercent values for fruit candy
fruit.ind  <- as.logical(candy$fruity)
fruit.candy <- candy[fruit.ind, ]
fruit.win   <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

On average, chocolate candy is higher ranked than fruity candy as the mean winpercent value for chocolate candy is higher than for fruity candy.

Q12. Is this difference statistically significant?

```
# Use a t-test to analyze statistical significant of the difference between winpercent values
t.test(choc.win, fruit.win)
```


Welch Two Sample t-test

```
data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, the difference is statistically significant as the p-value is small, 2.871e-08, which was found by using the function `t.test()`.

Overall Candy Rankings

Use the base R `order()` function together with `head()` to sort the whole dataset by a variable/column of interest.

Q13. What are the five least liked candy types in this set?

The five least liked candy types in this dataset are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```
# Use the `order()` function with `head()` to sort the candy dataset by winpercent and set n
head(candy[order(candy$winpercent), ], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197	0.976	
Boston Baked Beans				0	0	0	1	0.313	0.511	
Chiclets				0	0	0	1	0.046	0.325	
Super Bubble				0	0	0	0	0.162	0.116	
Jawbusters				0	1	0	1	0.093	0.511	

winpercent

Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
# Use the `order()` function with `head()` to sort the candy dataset by winpercent and set n
# Set decreasing = TRUE to find the highest winpercent values
head(candy[order(candy$winpercent, decreasing = TRUE), ], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0		0		0.720
Reese's Miniatures		0	0	0		0		0.034
Twix		1	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Snickers		0	0	1		0		0.546

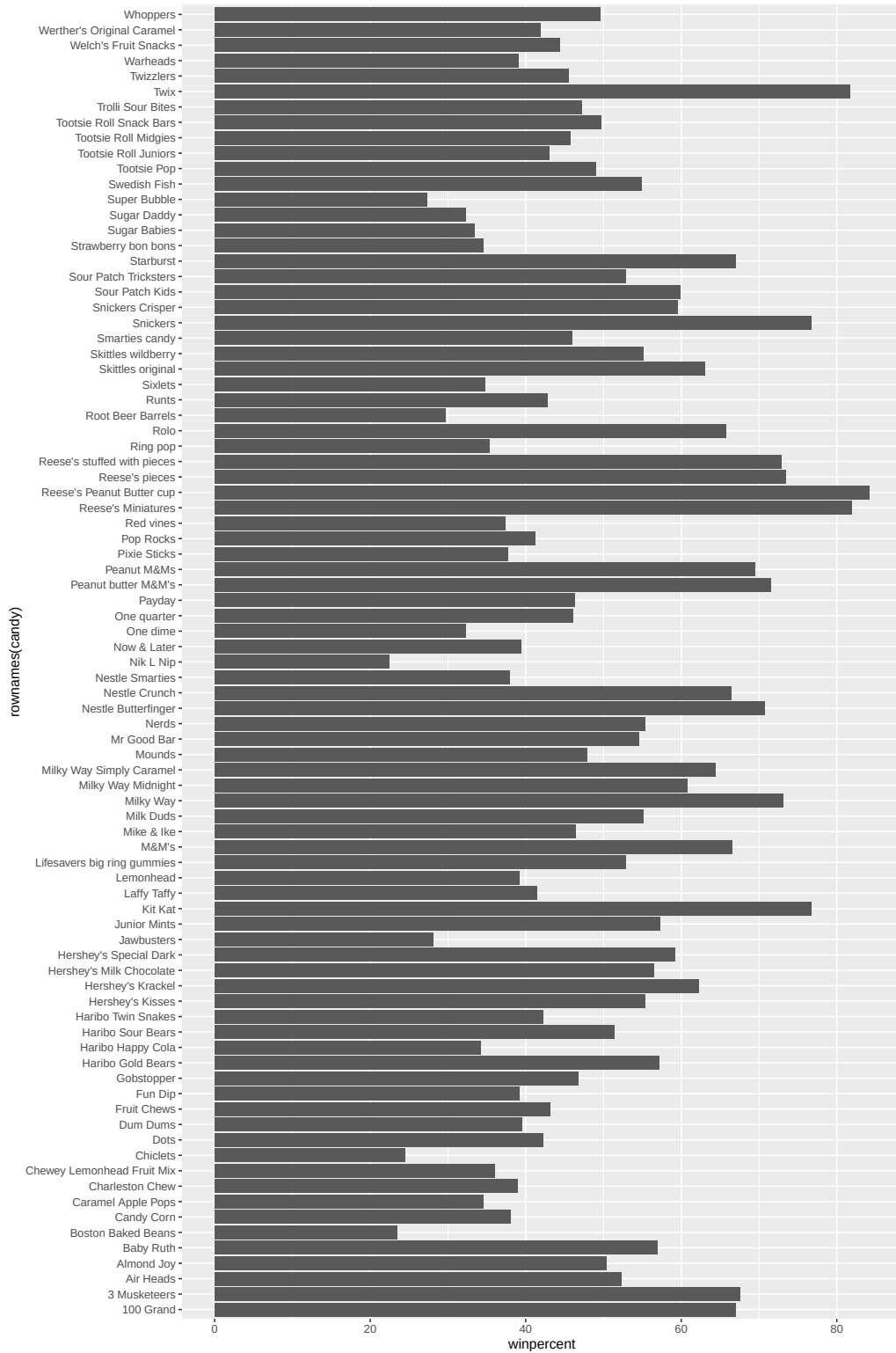
	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

The top 5 favorite candy types are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

```
# Load ggplot2
library(ggplot2)

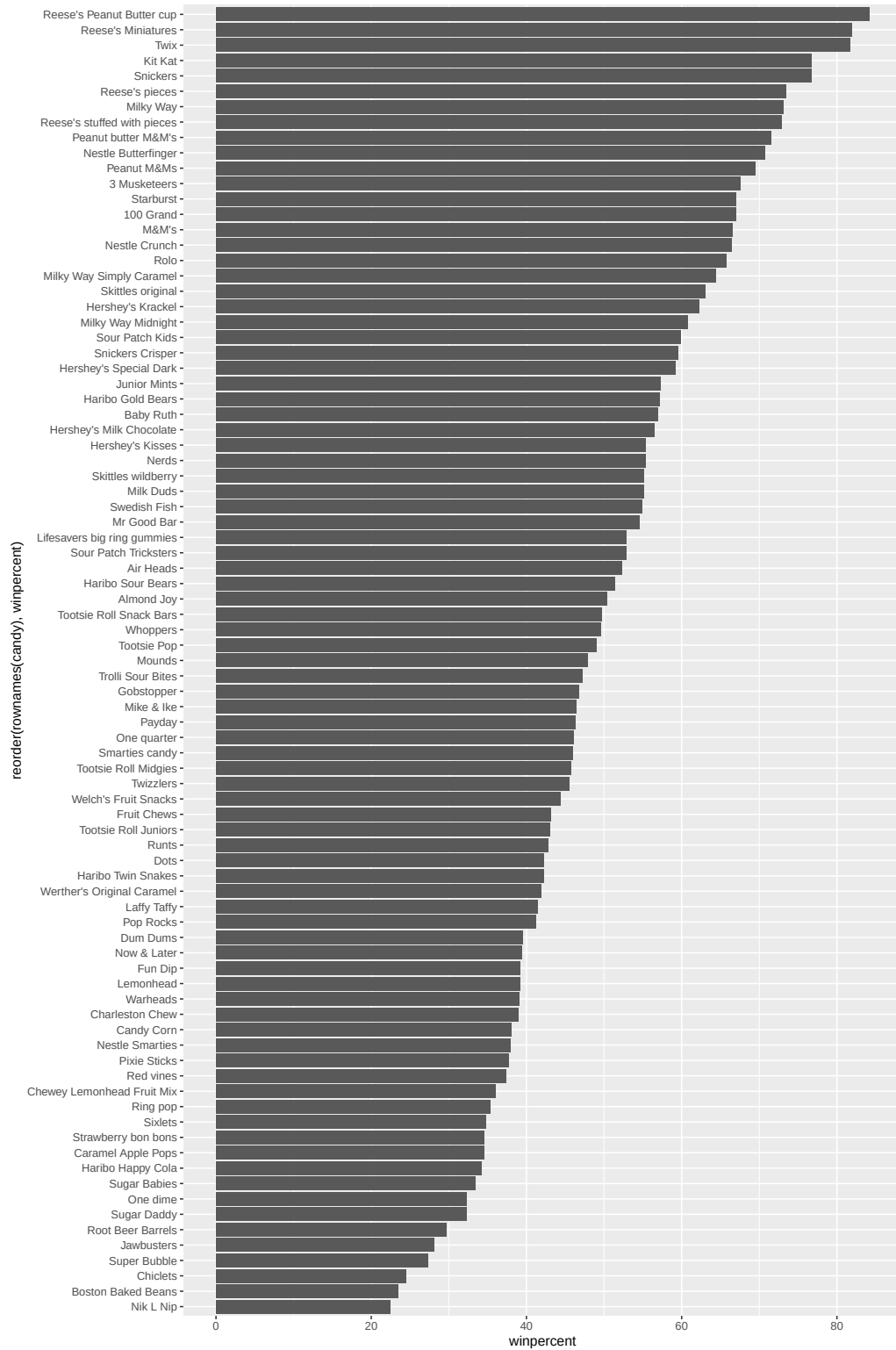
# Make a first barplot of candy ranking based on winpercent values using ggplot
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
# Barplot with candies sorted by winpercent

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



Time to add some useful color

Let's setup a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

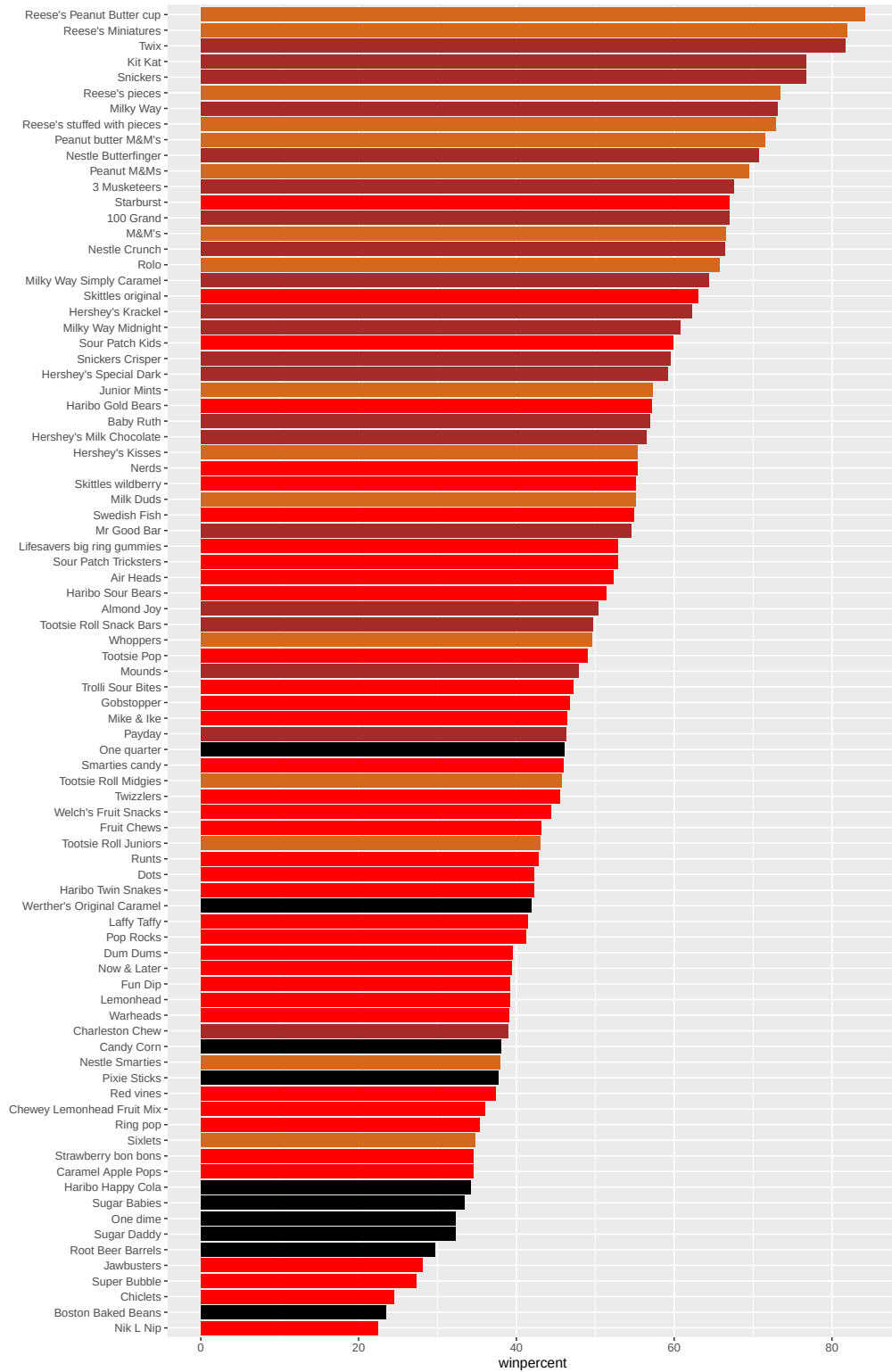
```
# Set up color vector of all gray values (one for each candy)
my_cols <- rep("black", nrow(candy))
```

```
# Overwrite chocolate for chocolate candy
my_cols[as.logical(candy$chocolate)] <- "chocolate"
```

```
# Overwrite brown for candy bars
my_cols[as.logical(candy$bar)] <- "brown"
```

```
# Overwrite red for fruity candy
my_cols[as.logical(candy$fruity)] <- "red"
```

```
# Make barplot with this color scheme
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is “Sixlets” as it is the lowest chocolate candy on the barplot and has the lowest winpercent for chocolate candies.

Q18. What is the best ranked fruity candy?

The best ranked chocolate candy is “Starburst” as it is highest on the barplot for fruity candy and has the highest winpercent for fruity candy.

Taking a look at pricepercent

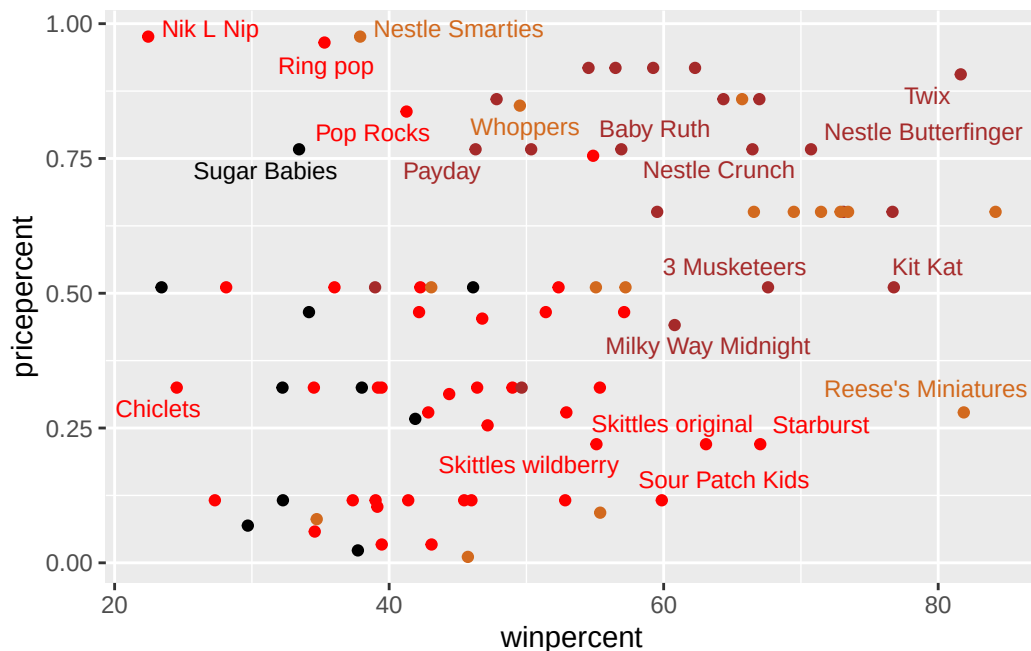
What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of winpercent vs the pricepercent variable. The pricepercent variable records the percentile rank of the candy’s price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

To this plot we will add text labels so we can more easily identify a given candy. There is a regular `geom_label()` that comes with `ggplot2`. However, as there are quite a few candies in our dataset lots of these labels will be overlapping and hard to read. To help with this we can use the `geom_text_repel()` function from the *ggrepel* package.

```
#install.packages("ggrepel")

# Load ggrepel package
library(ggrepel)

# Plot winpercent vs pricepercent
ggplot(candy) +
  aes(winpercent, pricepercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```

Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
head(candy[order(candy$winpercent/candy$pricepercent, decreasing=TRUE), c(11,12)], n=1)
```

	pricepercent	winpercent
Tootsie Roll Midgies	0.011	45.73675

The candy type that is the highest ranked in terms of winpercent for the least money is Tootsie Roll Midgies.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
# To see which candy is the most expensive we can order() the dataset by pricepercent.
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076

Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive candies are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular is Nik L Nip as it has the lowest winpercent value.

```
#Using the dplyr package we can do the same thing

library(dplyr)
candy |>
  arrange(-pricepercent) |>
  select(pricepercent, winpercent) |>
  head(n=5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the correlation structure

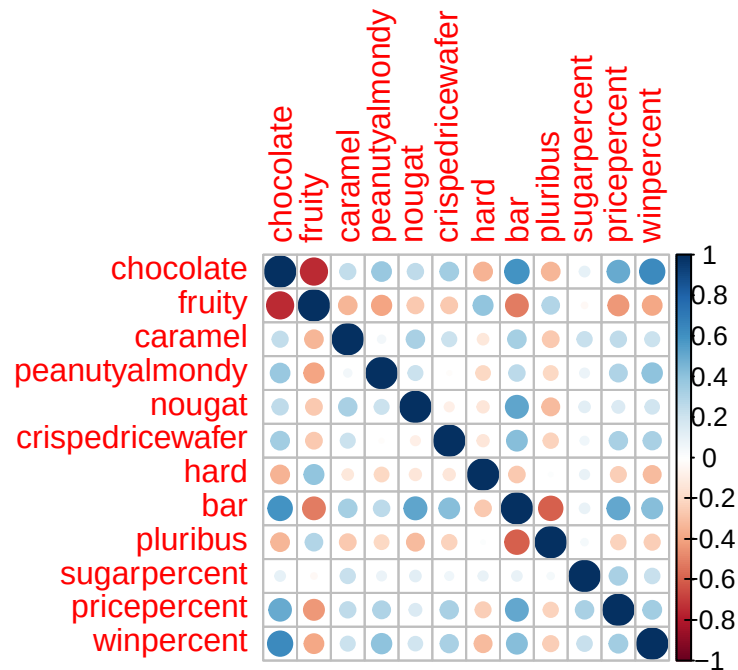
We can calculate the pair-wise correlation of all our columns

```
#install.packages("corrplot")
```

```
# Import corrplot package
library(corrplot)
```

corrplot 0.95 loaded

```
# Calculate correlation between all variables and view results in correlation plot
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are two variables that are anti-correlated because the candies tend to either be fruity or chocolate and not both.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

The variables I expect to have the largest contributions to PC1 are chocolate, fruity, bar, pricepercent, and winpercent variables. These variables drive the most separation between candies along the first principal component.

Principal Component Analysis

Apply PCA using the `prcomp()` function to our candy dataset remembering to set the `scale=TRUE` argument because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

```
# Apply PCA on candy dataset and scale variables so each variable equally contributes to PCA
pca <- prcomp(candy, scale = TRUE)
```

```
# Create summary of results
summary(pca)
```

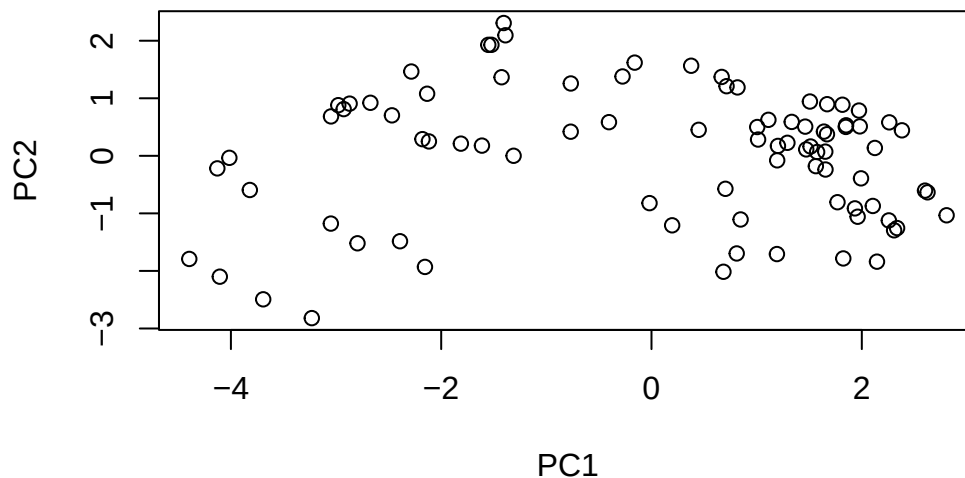
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

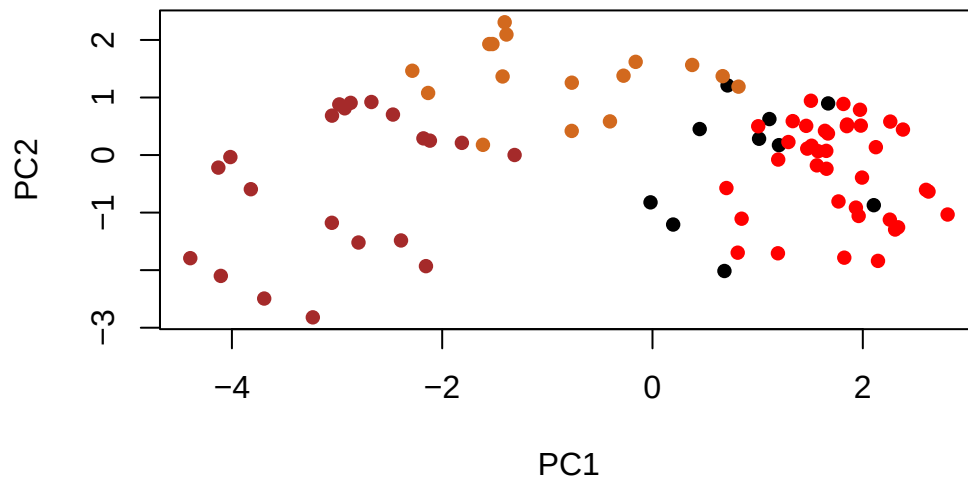
	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First major result figure is the “score plot” of PC1 vs. PC2 - how the different candy are related to each other on our new PC axis:

```
# Plot PC1 vs PC2 scores
plot(pca$x[, 1:2])
```



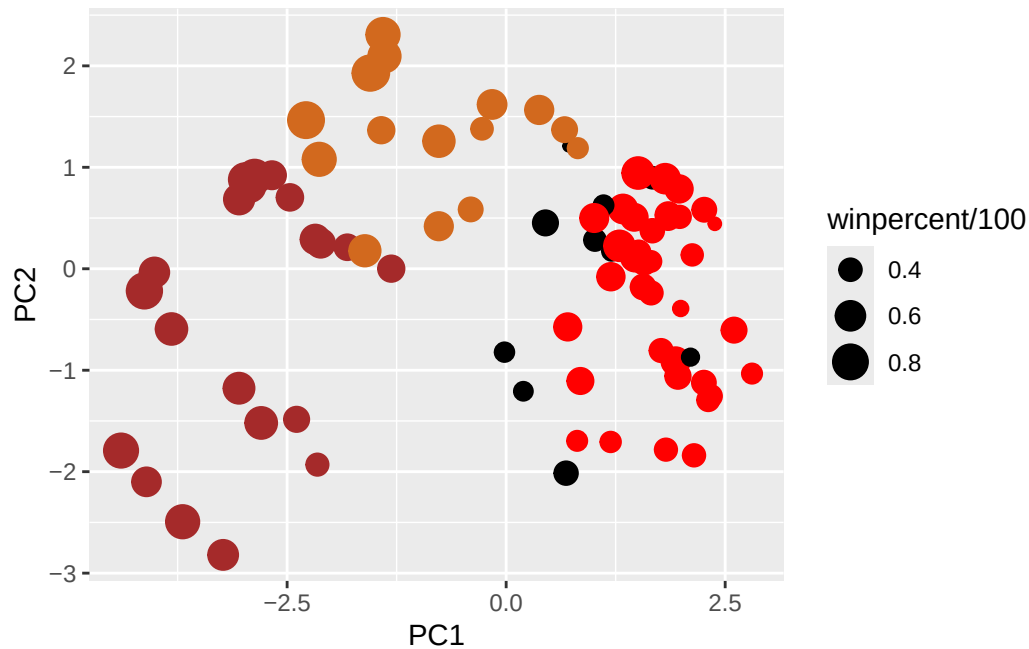
```
# Change the plotting character and apply my_cols
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data  
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +  
  aes(x=PC1, y=PC2,  
      size=winpercent/100,  
      text=rownames(my_data),  
      label=rownames(my_data)) +  
  geom_point(col=my_cols)
```

p



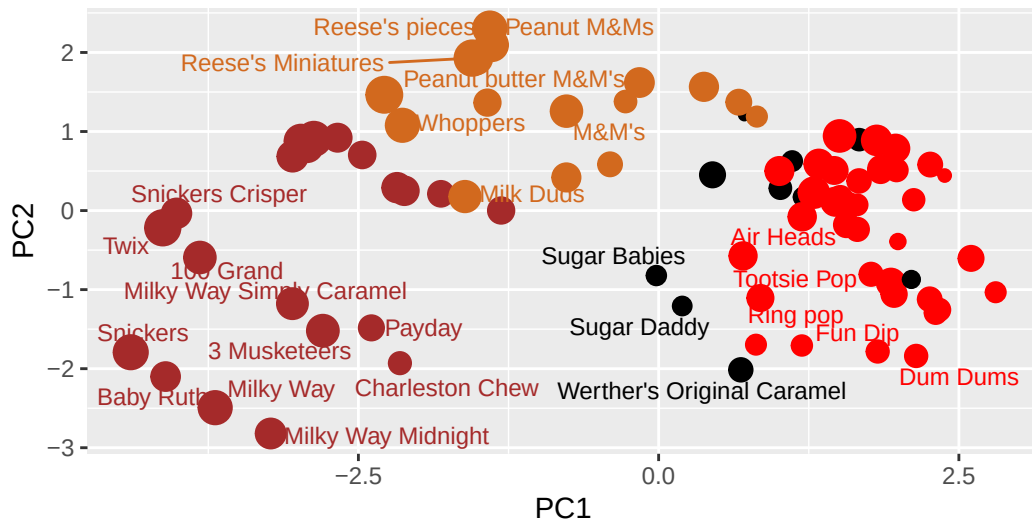
Use the ggrepel package and the function `ggrepel::geom_text_repel()` to label up the plot with non overlapping candy names like. Also add a title and subtitle

```
library(ggrepel)

#label up the plot with non overlapping candy names
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
        caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
#install.packages("plotly")
```

```
# Use "plotly" package
```

```
#library(plotly)
```

```
# Generate interactive plot to see all labels by hovering over each point
```

```
#ggplotly(p)
```

The second major results from PCA is the so-called “loadings plot”

Let’s finish by taking a quick look at PCA our loadings. Do these make sense to you? Notice the opposite effects of chocolate and fruity and the similar effects of chocolate and bar

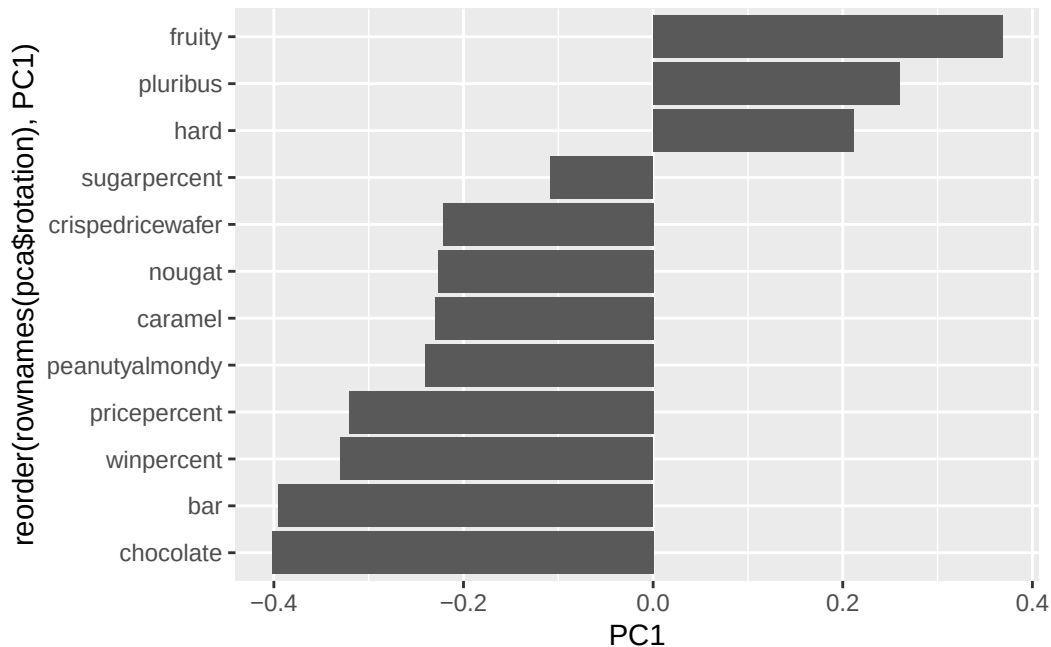
```
# Generate a PCA loadings plot with PC1 vs PC2
```

```
ggplot(pca$rotation) +
```

```
  aes(PC1,
```

```
      reorder (rownames(pca$rotation), PC1))+
```

```
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables picked up strongly by PC1 in the positive direction are fruity, pluribus, and hard. This makes sense to me because fruity candy tends to be sold in multiple pieces and are a lot of times hard candy. This is different than chocolate candies which tend to have nougat, caramel, peanut or almond, and have higher winpercent. This relationship was previously highlighted in the corplot that showed the anti-correlation between chocolate and fruity candy. The candy with factors relating to fruity candy tend to not be chocolatey, and vice versa.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The combination of a chocolate candy that is not fruity and is in the form of a bar appears to make a “winning” candy. Also having the variables associated with chocolate candy like peanut, almond, or caramel leads to a desirable candy. The visualization, correlation, and PCA analysis support and complement each other in reaching this conclusion as visualizing the data showed the most popular individual candy, which tends to be chocolate. This data analysis works together with the corplot to show correlation between different candy variables. Knowing chocolate candy has the highest winpercent values, we see variables correlated with and anti-correlated with it. This allowed us to see that chocolate and fruity candy are negatively

correlated, so fruity candy can be concluded to be less desirable. Finally, PCA contributes to this analysis by showing the division between the candy types, confirming the winning variables.